How Visual Cues Shape Happiness Perception: The Roles of Expression and Colour

Abstract

Facial expressions and colour cues both shape how observers interpret emotion, yet their combined influence is understudied. This study examined how smile morphology (open-mouth vs. neutral) and background colour (yellow vs. red) jointly affect happiness judgements. Fifty-two participants completed a 2 × 2 within-subjects task, rating faces displaying either an open-mouth smile or a neutral expression against yellow or red backgrounds. Open-mouth smiles were rated much happier than neutral expressions, replicating established effects of smile intensity. Background colour showed a small, non-significant main effect, with slightly higher ratings for yellow. Crucially, a significant interaction revealed that the difference between open-mouth and neutral expressions was larger on red than yellow backgrounds, indicating that colour altered the strength of the facial signal. These findings support models proposing that facial features and contextual cues are integrated during emotion perception and highlight implications for colour-emotion associations and the interpretation of ambiguous expressions.

Introduction

Human facial expressions play a key role in social communication, helping people quickly interpret others emotions and intentions. Evolutionary accounts propose that expressions function as adaptive signalling systems, shaped by natural selection to address recurring social challenges such as threat detection, affiliation, and appeasement (Darwin, 1872). Within this system, the smile is one of the most socially significant expressions. Early ethological research distinguished between low-intensity, closed-mouth smiles - often interpreted as appeasement or polite displays (Preuschoft, 1992; Van Hoof, 1972) - and higher-intensity, open-mouth smiles that signal genuine positive affect. Contemporary

research strongly supports this distinction: open-mouth smiles are consistently perceived as more intense, more genuine, and more strongly associated with happiness than closed-mouth or neutral expressions (Krumhuber & Manstead, 2009; Ambadar et al., 2009; Nusseck & Cunningham, 2014). These findings highlight that smile morphology is a powerful cue for judging emotional valence.

However, facial expressions are rarely interpreted on their own. Increasing research demonstrates that contextual cues, including colour, shape how observers perceive emotional meaning. Colour-emotion associations are robust across cultures: yellow is reliably linked to happiness, warmth, and approach motivation (Kaya & Epps, 2004; Bockelman et al., 2013; Valdez & Mehrabian, 1994; Ou et al., 2004), whereas red is associated with avoidance, threat, and negative affect (Elliot & Maier, 2014). Experimental research shows that red backgrounds can bias observers to perceive anger or hostility in otherwise ambiguous faces (Young et al., 2013; Fetterman et al., 2014). In contrast, yellow hues tend to encourage more positive interpretations and can enhance the processing of happy expressions (Jonauskaite et al., 2020). Together, these findings suggest that colour cues may act as emotional cues, subtly shifting the perceived emotional meaning of a face. Despite this growing evidence, the literature is still incomplete, both theoretically and empirically. Many studies look at colour or facial expressions in isolation, leaving unanswered questions about how these cues interact when combined. Those that do examine colour and facial emotion together often rely on highly saturated or artificial stimuli, which limits ecological validity, or focus on a single emotion rather than exploring variations in expression intensity. Crucially, very few studies have explored how colour cues influence the perceived happiness of different smile types, even though theory strongly predicts such an interaction. This gap highlights a clear tension in the literature: if both smile type and colour shape emotional interpretation, how are these cues combined when they send converging or conflicting signals?

The present study tackles this gap by examining how smile type (open-mouth vs. neutral) and background colour (yellow vs. red) jointly influence happiness judgments. Theoretical reasoning suggests not only main effects but also a meaningful interaction. Open-mouth smiles provide a clear, unambiguous signal of happiness, which yellow backgrounds may enhance and red backgrounds may diminish. For neutral expressions - where emotional meaning is less clear - colour cues may play a stronger role, with yellow pushing interpretations toward positivity and red toward negativity. This approach aligns with models of emotion perception that emphasise the integration of contextual cues and facial features, rather than treating them independently (Takei & Imaizumi, 2022; Thorstenson et al., 2018). Based on this literature, three hypotheses were proposed. **H1:** Open-mouth smiles will be rated as happier than neutral expressions. **H2:** Faces presented on yellow backgrounds will be rated as happier than those on red backgrounds. **H3 (Interaction):** The positive effect of an open-mouth smile will be stronger on yellow backgrounds and weaker on red backgrounds, reflecting the integration of facial and colour cues in emotion perception.

Method

Design

The study used a 2 × 2 within-subjects design with smile type (open-mouth vs. neutral) and background colour (yellow vs. red) as independent variables. The dependent variable, perceived happiness, was measured on a 0-100 slider, 0 being “not happy at all” and 100 “very happy”. All participants completed all conditions in a randomised order to minimise order, learning, and carry-over effects. This allowed examination of both main effects and the predicted interaction between smile and colour.

Participants

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Seventy-two participants were recruited through opportunity sampling. Twenty were excluded for excessively long completion times, leaving 52 participants (33 females, 19 males; age range 18-61, M = 31.70, SD = 13.50). All reported normal or corrected-to-normal vision. An a priori G*Power analysis indicated a minimum sample of 45 for detecting a medium within-subjects interaction with 80% power at $\alpha = .05$. All participants provided informed consent, were debriefed, and received ethical approval.

Materials

The facial stimuli consisted of multiple adult identities, each with one open-mouth smile and one neutral expression, selected from the Gorilla.sc facial-expression library because they met standard criteria for emotion-research stimuli (direct gaze, frontal orientation, neutral lighting). Following common practice in colour-emotion studies, the original backgrounds were removed and replaced with solid yellow or red backgrounds using fixed RGB values (Yellow: 255,255,0; Red: 255,0,0) to ensure consistent colour presentation. The experiment was hosted on Gorilla.sc. Participants rated each face on a 0-100 happiness slider (default midpoint) using a desktop or laptop and reported normal or corrected-to-normal vision.

Procedure

Participants were recruited through opportunity and snowball sampling. After opening the study link, they provided informed consent and completed brief demographic questions. They were instructed to rate “how happy the person looks” using a 0-100 slider. The experiment consisted of 54 trials presenting repeated combinations of expression (open-mouth vs neutral) and background colour (yellow vs red). Trials were fully randomised. On each trial, a face appeared centrally against either a yellow or red background, and participants submitted a rating before continuing. After completing all trials, participants were debriefed and informed of the study’s purpose. The procedure took approximately 3-5 minutes.

Results

Participants completed fifty-four within-subject conditions created by crossing background colour (red, yellow) with facial expression (happy, neutral). Descriptive statistics showed that happy faces received substantially higher ratings (M = 76.0, 95% CI [73.3, 78.7]) than neutral faces (M = 44.5, 95% CI [41.9, 47.1]). Ratings were also slightly higher for yellow backgrounds (M = 61.4, 95% CI [59.2, 63.5]) than red backgrounds (M = 59.1, 95% CI [57.0, 61.2]), suggesting a strong effect of expression and a smaller effect of colour.

Assumption checks were conducted prior to analysis. Visual inspection of the Q-Q plot indicated that residuals followed the expected normal distribution, and the scale–location plot showed no evidence of heteroscedasticity. Given the repeated-measures design and the relatively large number of observations (N = 208), the ANOVA was considered robust to minor deviations from normality. Diagnostic plots are provided in Appendix A.

A 2 x 2 repeated-measures ANOVA examined the effects of colour (red, yellow) and expression (happy, neutral) on happiness ratings. There was a significant main effect of colour, $F(1, 51) = 5.01$, $p = .030$, $\eta^2G = .01$, 95% CI [0.00, 0.13], with slightly higher ratings for yellow than red backgrounds. There was a very large main effect of expression, $F(1, 51) = 288.45$, $p < .001$, $\eta^2G = .70$, 95% CI [0.56, 0.79], indicating that happy faces were rated much higher than neutral faces. The colour × expression interaction was also significant, $F(1, 51) = 112.54$, $p < .001$, $\eta^2G = .06$, 95% CI [0.00, 0.22], demonstrating that the effect of expression differed across background colours.

Bonferroni-adjusted simple effects clarified the interaction. For happy faces, ratings were only slightly higher on yellow than red backgrounds (estimate = 2.75, $t(51) = 2.53$, $p = .015$). For neutral faces, ratings were substantially lower on red than yellow backgrounds (estimate

= -7.32, t(51) = -6.28, p < .001). The expression effect was significant within both colours but larger in the red condition (difference = 36.5, p < .001) than in the yellow condition (difference = 26.4, p < .001). Estimated marginal means are presented in Table 1, and Figure 1 illustrates the steeper decline from happy to neutral ratings in the red condition.

Table 1

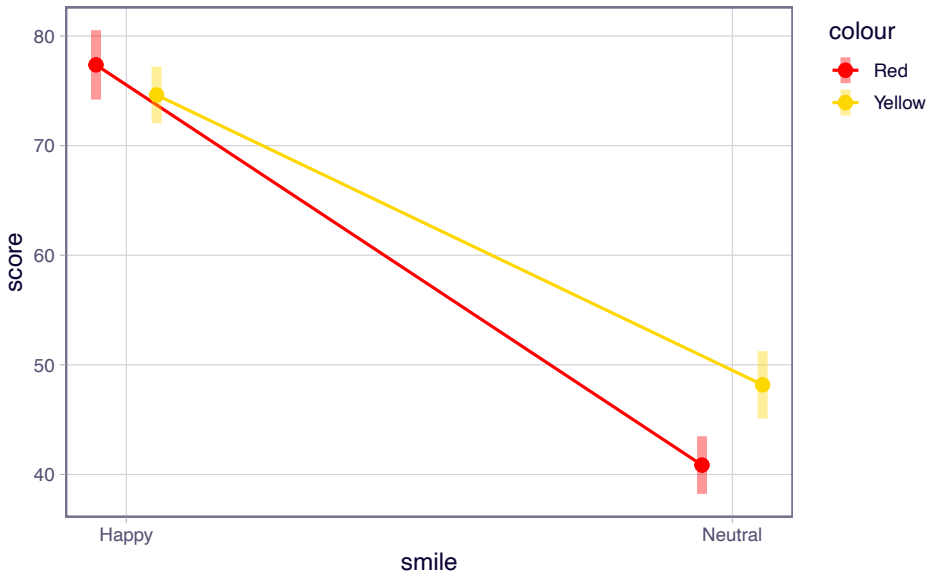
Table 1. Estimated Marginal Means for the 2 x 2 Repeated-Measures ANOVA

Colour	Smile	Mean	Lower CI	Upper CI
Red	Happy	76.0	73.3	78.7
Red	Neutral	44.5	41.9	47.1
Yellow	Happy	76.0	73.3	78.7
Yellow	Neutral	51.8	49.2	54.4

Figure 1

Interaction between background colour and facial expression on rating scores.

Error bars represent 95% confidence intervals.



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Discussion

The present study examined how smile morphology and background colour jointly influence happiness perception. As predicted, open-mouth smiles were rated as much happier than neutral expressions, echoing strong evidence that people rely heavily on how intense a smile is when judging emotion (Krumhuber & Manstead, 2009; Ambadar et al., 2009). This large effect aligns with evolutionary accounts suggesting that high-intensity smiles act as strong affiliative signals processed quickly and with confidence (Darwin, 1872; Preuschoft, 1992). The consistency of this finding highlights the especially powerful role that smiles play in social communication, where small differences in morphology can shape emotional interpretation (Nusseck & Cunningham, 2014).

A statistically significant but very small effect of background colour also emerged, with faces on yellow backgrounds receiving slightly higher happiness ratings than those on red backgrounds. The minimal effect size suggests that colour alone had only a subtle influence on emotional interpretation. This aligns with research showing that colour-emotion associations are reliable but often modest, particularly when the emotional signal is already clear (Elliot & Maier, 2014). Yellow is commonly associated with positivity and approach motivation (Kaya & Epps, 2004; Valdez & Mehrabian, 1994), whereas red is linked to avoidance and threat (Fetterman et al., 2014). These associations tend to have their strongest impact when the stimulus is ambiguous (Young et al., 2013), which helps explain why the overall colour effect was detectable but small.

The significant interaction between smile type and background colour provides the clearest theoretical insight. Colour had only a minimal influence on open-mouth smiles, but it had a much larger impact on neutral expressions, which were rated considerably less happy on red backgrounds than on yellow ones. This pattern supports cue-integration models of emotion

perception, which propose that observers combine facial features with contextual information, weighting each cue according to its reliability (Thorstenson et al., 2018; Takei & Imaizumi, 2022). When the expression was clear, the facial cue dominated and colour had little effect. When the expression was ambiguous, colour played a stronger role: yellow shifted interpretations toward positivity, whereas red reduced perceived happiness. This mirrors findings from multimodal emotion research showing that contextual cues have their greatest influence when primary emotional information is weak or uncertain (Aviezer et al., 2017).

A methodological strength of the study is the use of multiple facial identities. Perceptions of emotion can vary depending on characteristics such as attractiveness, age, or facial structure (Todorov et al., 2015). Including a diverse range of faces reduces identity-specific bias and improves the generalisability of the interaction between expression and colour. A second strength is the within-subjects design, which ensured that each participant saw all combinations of colour and expression. This controls for individual differences in rating style and increases statistical power, making it well suited for detecting subtle contextual influences such as colour effects.

Several limitations should be considered. Online testing can introduce variability in screen calibration and ambient lighting, both of which may influence colour perception. Replicating the study in a laboratory setting, or incorporating device-standardisation checks, would increase confidence in the colour effects observed. The use of opportunity and snowball sampling may also limit demographic diversity, and cultural differences in colour-emotion associations are well documented (Jonauskaite et al., 2020). Cross-cultural replications would therefore be valuable. Finally, the study focused only on happiness; future research could examine whether similar interactions occur for other emotions, particularly those with more subtle or ambiguous expressions.

Overall, the present study shows that smile morphology and background colour interact to shape perceptions of happiness. Although facial expression is still the main thing people rely on, colour can shape how a face is interpreted when the emotion is less clear. These findings support integrative models of emotion perception and highlight the important role of contextual cues in shaping how observers interpret emotional meaning from faces.

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Appendix

Appendix A

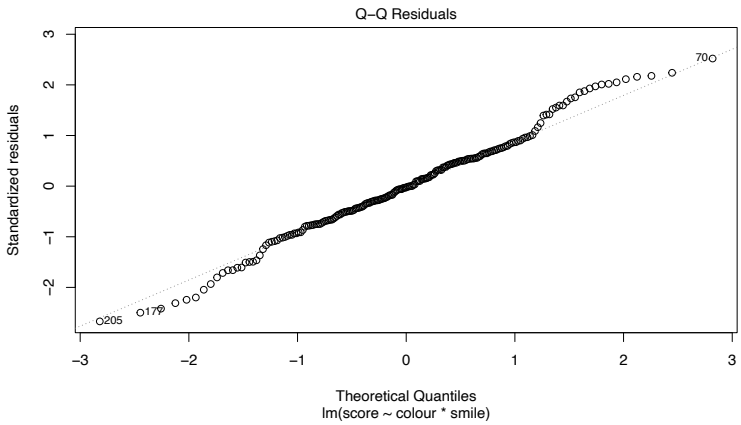
Assumption Checks

A.1 Q-Q Plot of Standardised Residuals

Figure A1

Q-Q plot showing the distribution of residuals from the repeated-measures ANOVA model.

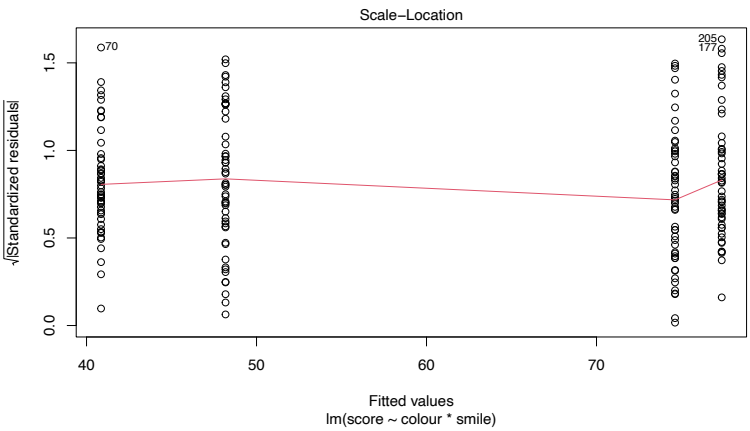
Points fall close to the diagonal line, indicating acceptable normality.



A2. Scale–Location Plot

Figure A2

Scale-location plot showing the spread of residuals across fitted values. Residuals appear evenly distributed, indicating no evidence of heteroscedasticity.



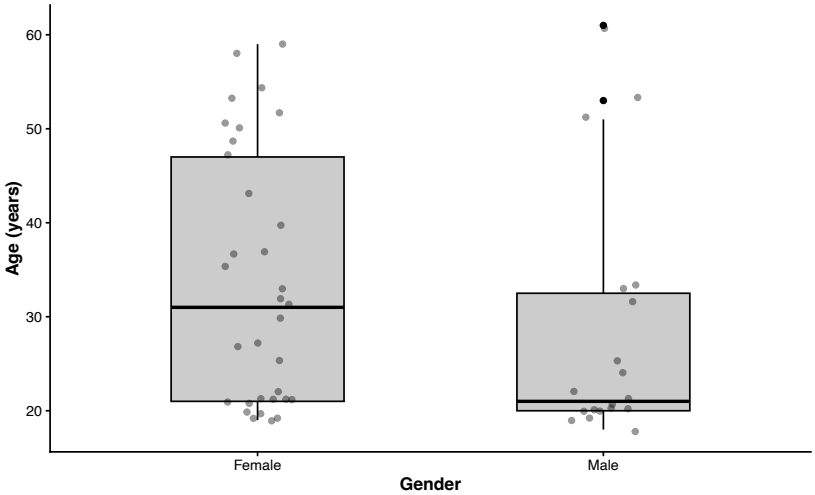
Appendix B

Exploratory Data Visualisation

B1. Boxplot of Age by Gender

Figure B1

Exploratory boxplot used during initial data screening. This plot was not part of the inferential analysis.



Appendix C

Script of AI Assistance Conversation

The following is a transcript of a conversation between the author and an AI assistant used for guidance on statistical concepts, research methods, and academic writing.

Author: What does Homoscedasticity mean? **AI Assistant:** Homoscedasticity means that the variance of the errors or residuals is constant across all levels of an independent variable in a statistical model.

Author: Give this in a sentence easy to understand. **AI Assistant:** Homoscedasticity means that the spread of the errors in a model stays roughly the same for all values of the data.

Author: I don't understand what I've done. This is my R output: Error in -pwr.f2.test(u = 1, f2 = 0.15, power = 0.8, sig.level = 0.05): invalid argument to unary operator. **AI Assistant:** The error occurs because of the minus sign at the start of your command. Remove it:

pwr.f2.test(u = 1, f2 = 0.15, power = 0.8, sig.level = 0.05)

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Author: I don't understand this one: Error in **group_by()**: Column **id** is not found. Column **colour** is not found. Column **smile**is not found.**AI Assistant:** This means the columns you tried to group by do not exist in your dataset. Check the column names using **names(your_dataframe)** and correct them.

Author: What is a Q-Q plot? **AI Assistant:** A Q-Q plot is a graph used to check if a dataset follows a normal distribution by comparing the data's quantiles to the expected quantiles of a normal distribution.

Author: What is estimated marginal mean? **AI Assistant:** Estimated marginal means are the adjusted group averages predicted from a statistical model after accounting for other variables.

Author: "Yellow" for my interaction plot doesn't show up very yellow. What other colour would show up stronger? **AI Assistant:** Use **"gold"**, **"goldenrod"**, or **"orange"** in R for stronger visibility. **"Gold"** is the most common choice.

Author: What are models of emotion perception?**AI Assistant:** Models of emotion perception are theories explaining how people recognize and interpret others' emotions from cues like facial expressions, voice, and context.

Author: What's learning effects? **AI Assistant:** Learning effects are improvements in performance caused by practice or familiarity with a task over time.

Author: Paragraph openers. **AI Assistant:** Examples include: "One key aspect is…", "Additionally…", "For example…", "In contrast…", "As a result…", "In conclusion…".

Author: However synonym. **AI Assistant:** Synonyms include: nevertheless, nonetheless, even so, that said, on the other hand, yet.

Author: Word for make sense of. **AI Assistant:** Alternatives include: understand, grasp, comprehend, interpret, decipher, figure out.

Author: Although synonyms. **AI Assistant:** Alternatives include: even though, though, while, whereas, albeit, notwithstanding the fact that.

Author: What do F and t mean when writing results? **AI Assistant:** F is the statistic from ANOVA comparing three or more group means. t is the statistic from a t-test comparing two group means. Both are reported with degrees of freedom and p-values.

Author: What goes first in APA7, table or figure? **AI Assistant:** Tables go first, followed by figures.

Author: For my scale location plot, the red line is fairly flat and no strong funnel shape, however there is slight curvature at far right. Is that okay? **AI Assistant:** Yes, that is generally okay. Minor curvature at extremes usually does not violate homoscedasticity.

Author: Roles smiles play in social communication. A**I Assistant:** Smiles express emotion, regulate social interaction, build relationships, and positively influence how others perceive us.

Author: Can opportunity and snowball sampling limit demographic diversity? **AI Assistant:** Yes. Opportunity sampling may favor easily reachable participants, and snowball sampling often produces clusters of similar demographics, limiting diversity.

Author: How many references in an experimental report should I have? **AI Assistant:** For a typical undergraduate report, 10–20 relevant references are common.

Author: Is 20 too much? **AI Assistant:** No, 20 references are reasonable if they are all relevant and supportive of your report.

Author: Is “overall” or “in conclusion” better to end a discussion? **AI Assistant:** “In conclusion” is more formal and traditional; “Overall” is slightly less formal and emphasizes a general summary.

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Author: Contemporary research strongly supports this distinction… Is this sentence too long?**AI Assistant:** Slightly long; consider breaking it into two sentences or using a bullet/list for clarity.

Author: Synonym for cues. **AI Assistant:** Alternatives include signals, signs, indicators, clues, pointers, gestures, expressions, body language, prompts, or stimuli.

Author: What do I put in an appendix? **AI Assistant:** Supplementary material such as raw data, detailed calculations, questionnaires, additional figures/tables, instructions/scripts, or ethics documents.

Author: Create a script of our conversation to put in my appendix APA7 style. **AI Assistant:** [This transcript, formatted for APA 7].